



Zoo Conservation × Education Mission

On-Chain Realization of
the 2021 Foundation Vision

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Abstract

The October 31, 2021 Zoo whitepaper established the Zoo Foundation as the conservation × education arm of the Zoo ecosystem. Four years later, this paper documents the on-chain realization: the protocol-level sustainability tax, treasury custody under M-Chain MPC, conservation grant cycles via the DAO, the V1 species catalogue with their named non-profit partners, and the post-quantum-secure settlement layer that anchors every grant disbursement. We avoid restating the vision; instead, we list what shipped, when, and against which 2021 commitment. This paper is the canonical conservation-pillar artifact for 2025-12-15.

1 Mission

The Zoo Foundation was established to harness blockchain technology and the global community it connects to help preserve the world’s most critically endangered animal species. The mission was articulated in §7 of the October 31, 2021 founder whitepaper [1]. This paper documents what shipped against that mission.

1.1 Operational Definition

A 501(c)(3) non-profit organization (Zoo Labs Foundation Inc.) that:

- Maintains a digital twin (NFT-bound AI companion) for every species in its V1 catalogue.
- Routes a 1–3% sustainability tax on every protocol transaction to a treasury multisig governed by the Zoo DAO.
- Disburses treasury funds to named conservation partners through quarterly grant cycles.
- Publishes verifiable impact reports under the Conservation Impact Measurement framework (ZIP-0501) and the Evidence Locker Index (ZIP-0560).
- Operates a sanctuary visit and education program tied to in-situ field operations.

1.2 2021 Commitment vs. 2025-12-15 Realization

2021 Commitment (§7)	2025-12-15 Realization
Digital twins of endangered species	V1 NFT catalogue across 8 species (§2); each NFT bears an empathetic-AI avatar (ZIP-0400, ZIP-0402).
Monetary contribution to conservation partners	Sustainability tax (<code>contracts/specs/zoo-sustainability-tax.md</code>); routed to Foundation Treasury Multisig (M-Chain MPC, LP-019).
Education for all ages worldwide	Educational AI Pillar (ZIP-0008 avatar tutors; ZIP-0405 conservation-aware language models).
Fundraising events with partners	Quarterly grant cycles and Zoo Foundation events (off-chain, recorded on the Evidence Locker, ZIP-0560).
Mitigate illegal wildlife trade	Anti-poaching alert network (ZIP-0503); satellite ecological monitoring (ZIP-0425).
Sustainable animal sanctuaries	Habitat-restoration bonds (ZIP-0113); habitat conservation framework (ZIP-0520).
Direct-save monthly contribution	Micro-donation streaming (ZIP-0112); gasless conservation transactions (ZIP-0029).

The treasury is custodial under threshold-key signing on M-Chain (LP-019, MPC ceremonies)

and settles under QUASAR 3.0 (LP-020). Every grant disbursement is post-quantum-secure, end-to-end auditable, and lineage-traceable to a specific 2021 whitepaper commitment.

2 V1 Species Catalogue and Conservation Partners

The 2021 whitepaper enumerated eight V1 species (§18) with associated non-profit partners. The 2025-12-15 catalogue retains those species as foundational and adds Generation 1 candidates through the DAO (ZIP-0030, On-Chain Species Registry).

2.1 V1 Catalogue (2021 Genesis Cohort)

Species (Common)	Scientific	2025-12-15 Partner Set
Red Wolf	<i>Canis rufus</i>	Wolf Conservation Center, Defenders of Wildlife.
Amur Leopard	<i>Panthera pardus orientalis</i>	WWF Russia, ALTA (Amur Leopard & Tiger Alliance).
Nubian Giraffe	<i>Giraffa camelopardalis camelopardalis</i>	Giraffe Conservation Foundation.
Sumatran Elephant	<i>Elephas maximus sumatranus</i>	WWF Indonesia, Sumatran Elephant Conservation Initiative.
Pygmy Hippopotamus	<i>Choeropsis liberiensis</i>	ZSL West Africa Programme, IUCN SSC Pygmy Hippo Specialist Group.
Javan Rhinoceros	<i>Rhinoceros sondaicus</i>	International Rhino Foundation, Save the Rhino.
Siberian (Amur) Tiger	<i>Panthera tigris altaica</i>	WCS Russia, Panthera.
Emperor Penguin	<i>Aptenodytes forsteri</i>	British Antarctic Survey programmes; Pew Antarctic Project.

Table 1: V1 NFT catalogue with current partner set. Partner status is reviewed quarterly by the DAO under ZIP-0530 (Community Partnerships, FPIC).

Every catalogue species has a corresponding empathetic AI companion (ZIP-0402) and a Habitat NFT for fractional conservation funding (ZIP-0203). Generation 1 candidates — additional endangered species nominated by the community — are admitted through the On-Chain Species Registry (ZIP-0030) and require a Tier 2 non-profit partnership commitment.

2.2 Selection Criteria

1. IUCN Red List category Vulnerable, Endangered, or Critically Endangered.
2. At least one named non-profit field operation partner with established track record.
3. No legal blockers to grant disbursement in partner jurisdiction.
4. Available genomic / population data for the Conservation AI Pillar (ZIP-0500 series).

2.3 Multilateral Alignment

The catalogue and grant cycles align with multilateral conservation frameworks documented in ZIP-0550 (Conservation Standards Alignment): IUCN Red List, CITES Appendices, and the Global Biodiversity Framework targets. Compliance with each framework is on-chain attested through A-Chain (LP-134).

3 Treasury and Grant Cycles

3.1 Custody

The Zoo Foundation Treasury is held under M-Chain MPC threshold custody (Lux LP-019) with a 2-of-3 split between board signatories, the Foundation operations multisig, and a community-elected DAO representative. Every disbursement is co-signed and emits an on-chain receipt anchored on Q-Chain (QUASAR 3.0, LP-020).

3.2 Funding Sources

Treasury inflows come from four channels:

1. **Sustainability tax** — 1–3% on every protocol transaction (see `contracts/specs/zoo-sustainability-tax.mo`).
2. **Direct micro-donations** — streaming via ZIP-0112.
3. **Conservation bonds** — yield-bearing tokenized donations (ZIP-0101).
4. **Validator-rewards conservation split** — a percentage of validator rewards (ZIP-0019).

3.3 Grant Cycle (Quarterly)

Each cycle follows the framework in ZIP-0023 (Community Grant Program) and ZIP-0104 (Research Funding DAO Treasury):

1. **Open call.** Proposals submitted on-chain; gated by DID Tier (ZIP-0033) and minimum reporting commitment.
2. **Review.** Domain reviewers (catalogue partners, in-situ ecologists, citizen-science contributors) score proposals. Reviewers’ DIDs are published; reputation drift is bounded by ZIP-0026.
3. **Vote.** Weighted vote per the 2021 paper §28 formula: $\text{weight}(p) = \alpha \cdot a_p + \beta \cdot i_p + \gamma \cdot c_p + \delta \cdot s_p$ with stake capped at 10% of weight.
4. **Disbursement.** M-Chain co-signs payment; on-chain receipt anchors evidence to the Evidence Locker (ZIP-0560).
5. **Reporting.** Recipients commit to the Reproducibility Attestation framework (ZIP-0606) and a 90-day field-results report.

3.4 Allocation Limits

- Operations expense capped at 15% of treasury per cycle (the remainder must be programmatic or in-situ).
- Per-grant cap of 10% of treasury (avoids single-project concentration).
- Reserve floor of 25% retained against multi-year obligations.

The DAO can override these limits only by supermajority vote under ZIP-0017 (Treasury Spend $\geq 60\%$, $\geq 10\%$ quorum, 72-hour timelock).

4 Impact Measurement

4.1 Framework

Every grant cycle’s outcome is measured under the Conservation Impact Measurement framework (ZIP-0501) and indexed in the Evidence Locker (ZIP-0560). Impact data is published as open biodiversity data (ZIP-0601, Open Biodiversity Database Standard).

4.2 Verifiable Metrics

Metric	Source	Verification
Population census delta	Field partner	Multi-observer attestation; satellite cross-check (ZIP-0425).
Habitat hectares restored	Reforestation log	Satellite verification (ZIP-0505).
Anti-poaching incidents prevented	Anti-poaching alert network	On-chain alert log (ZIP-0503).
Educational reach	Avatar-tutor sessions (ZIP-0008)	Session attestations on A-Chain (LP-134).
Citizen-science contributions	Citizen-science protocol (ZIP-0602)	Reproducibility attestation (ZIP-0606).

Table 2: Verifiable impact metrics. Every metric has a defined source, a defined verifier, and a defined chain anchor.

4.3 Annual Impact Report

The Foundation publishes an annual impact report each calendar year end. The 2025 edition is the first operating under QUASAR 3.0 and the first end-to-end-verifiable report in the Foundation’s history.

4.4 Open Data

All non-sensitive impact data is published under ZIP-0601 (Open Biodiversity Database Standard), licensed CC-BY for research reuse. Sensitive location data (e.g., poaching-vulnerable species coordinates) is published behind selective-disclosure attestations (Z-Chain, LP-134) gated by researcher DID tier.

4.5 Cross-References

- 2021 Whitepaper §5–7 (Sustainability, Non-Profits, Foundation).
- Companion paper: *Zoo Foundation Mission* (Pillars I/II/III).
- Companion paper: *Zoo Conservation AI*.
- ZIP-0500 series (Conservation).
- ZIP-0570 (Zoo Labs Impact Thesis).
- `contracts/specs/zoo-sustainability-tax.md`.
- Lux LP-019 (M-Chain), LP-020 (QUASAR 3.0), LP-134 (chain topology).

References

- [1] A. Worring, *Zoo: A Game-Fi Multiverse Ecosystem for the Preservation of Endangered Species*, Zoo Labs Foundation, October 31, 2021. Reconstructed 2025-12-15.